

B	Busy time
λ	Arrival rate = #arrivals/T
X	Output rate = completions /T
U	Utilisation = B / T
S	Mean service time =
j / 0	Index of a device / system
V(j)	Visit ratio to device j
D(j)	Service demand of device j
N	Average #jobs (in a “box”)
R	Avg response time (in a “box”)
Z	Mean thinking time
M	#interactive users

Utilisation Law	$U = S X$
Forced Flow Law	$V(j) = X(j)/X(0)$
Service demand Law	$D(j)=U(j)/X(0)$
Little’s Law	$N=X*R$
Interactive response	$M=X(0)*(Z+R)$

Value	Probability	
1	0.1	1×0.1
2	0.1	2×0.1
3	0.2	3×0.2
4	0.1	4×0.1
5	0.3	5×0.3
6	0.2	6×0.2
	mwan	Sum of the above

$\text{Sum}_x \{ x * \text{Prob}[\text{getting } x] \}$

where x runs over all possible event

	Meaning
λ	Arrival rate
μ	service rate Mean service time = $1 / \mu$
ρ	Utilisation
m	Number of servers
n	Number of holding slots

For M/M/1 queue:

$$\rho = \lambda / \mu$$

Procedure:

Draw a diagram with the states

Add arcs between states with transition rates

Derive flow balance equation for each state, i.e.

Rate of entering a state = Rate of leaving a state

Solve the equation for steady state probability

Sum ($k * \text{Prob}[k \text{ jobs}]$)

$$= 0 * (1-\rho) + 1 * (1-\rho)\rho + 2 * (1-\rho)\rho^2 + 3 * (1-\rho)\rho^3 + \dots$$

=

$$(1-\rho)[1 * \rho + 2 * \rho^2 + 3 * \rho^3 + \dots]$$

Use the formula on p.20.

$p = ?$. $q = ?$. $x = ?$.

For $0 \leq x < 1$,

$$p + x(p + q) + x^2(p + 2q) + x^3(p + 3q) + \dots = \frac{p}{1-x} + \frac{xq}{(1-x)^2}$$

	Meaning
λ	Arrival rate
μ	service rate Mean service time = $1 / \mu$
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For M/M/1 queue:

$$\rho = \lambda / \mu$$

$$\text{Need : } \lambda < \mu$$

Throughput

= min(offered load, maximum processing rate)

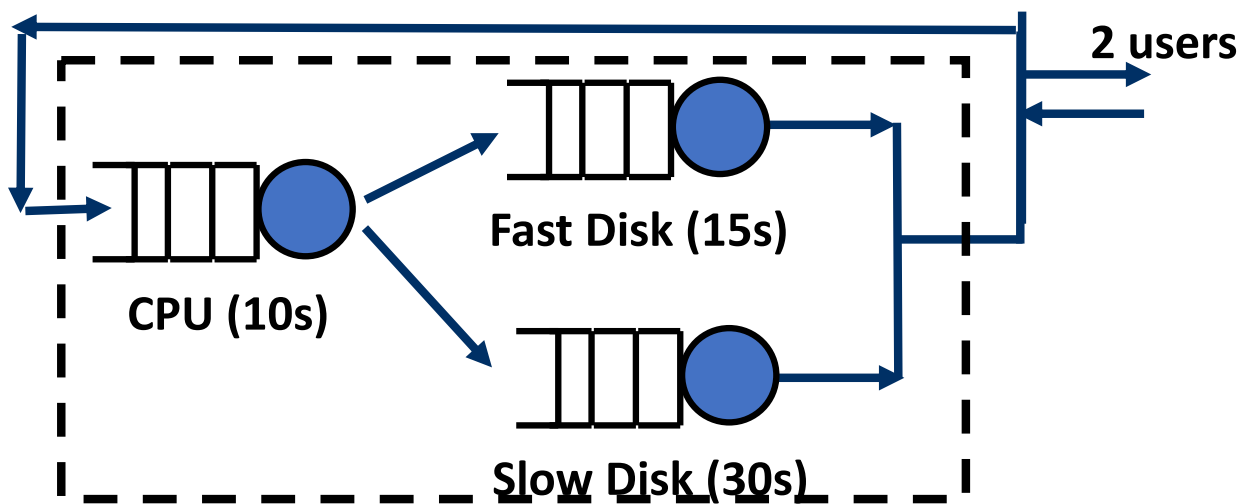
$$= \min(\lambda, \mu)$$

$$= \lambda$$

Probability [a job will finish its service in next δ seconds] = $\mu \delta$

Probability [a job will NOT finish its service in next δ seconds] = $1 - \mu \delta$

$$\begin{aligned} & \text{Prob}(\text{Job 1 will finish in the next } \delta \text{ seconds OR} \\ & \quad \text{Job 2 will finish in the next } \delta \text{ seconds}) \\ &= \text{Prob}(\text{Job 1 will finish in the next } \delta \text{ seconds}) + \\ & \quad \text{Prob}[\text{Job 2 will finish in the next } \delta \text{ seconds}] - \\ & \quad \text{Prob}[\text{Job 1 will finish in the next } \delta \text{ seconds AND} \\ & \quad \text{Job 2 will finish in the next } \delta \text{ seconds}] \\ &= \mu \delta + \mu \delta - (\mu \delta)^2 \\ &= 2 \mu \delta \end{aligned}$$



Choice 1:

(Location of User 1, Location of User 2) 3^n

Choice 2:

(#users at CPU, #users at fast disk, #users at slow disk)

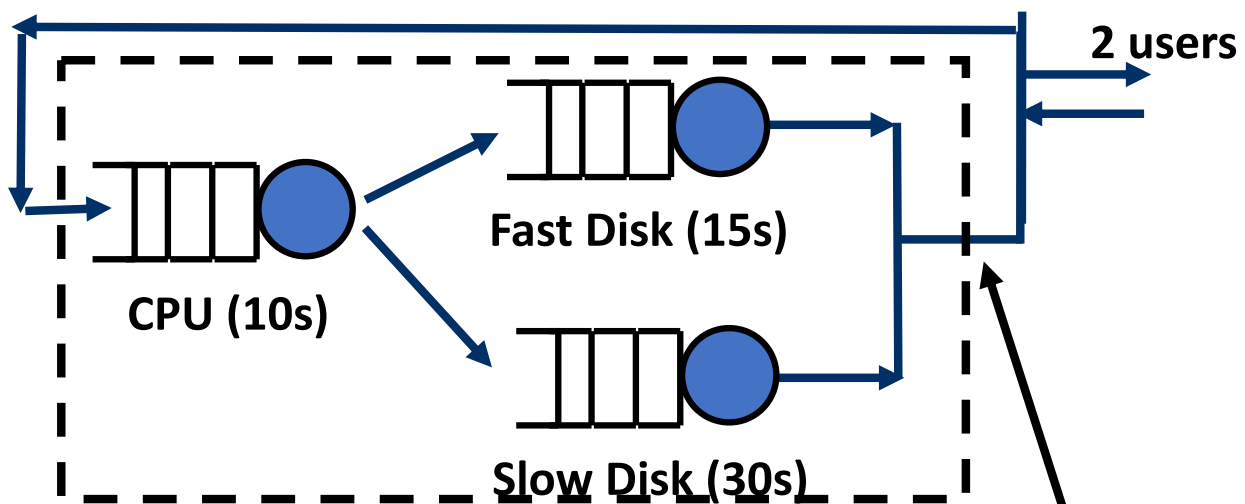
Comparing the two choices of states

(CPU, FD) (FD, CPU)

(1,1,0)

(2,0,0) $\rightarrow$ (0,1,1) (3 delta) * (3 delta)

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At steady state:
Throughput of the system
= Arrival rate of the CPU

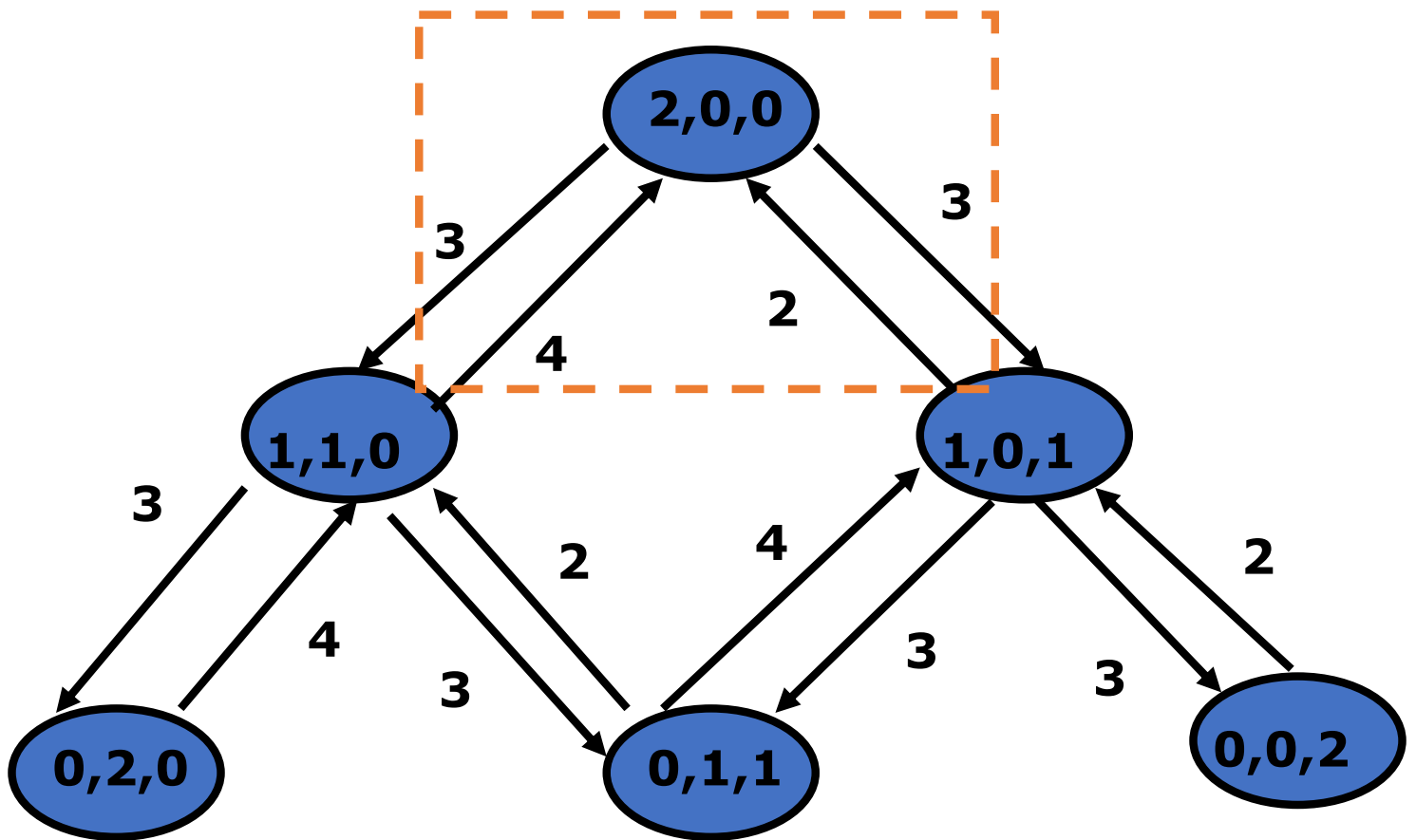
Arrival rate of CPU
= Throughput of the CPU

Water tank analogy

A pipe goes in, a pipe goes out

To have steady water level in the tank, you need

Rate of water into the tank = rate of water out of the tank



Rate of going into the state $(2,0,0)$
 $= 4 P_{110} + 2 P_{101}$

Rate of leaving the state $(2,0,0)$
 $= 3 P_{200} + 3 P_{200}$
 $= 6 P_{200}$

$$4 P_{110} + 2 P_{101} = 6 P_{200}$$

2 users

(2,0,0)

(1,1,0), (1,0,1)

(0,2,0), (0,1,1) (0,0,2)

4 users

(4,0,0)

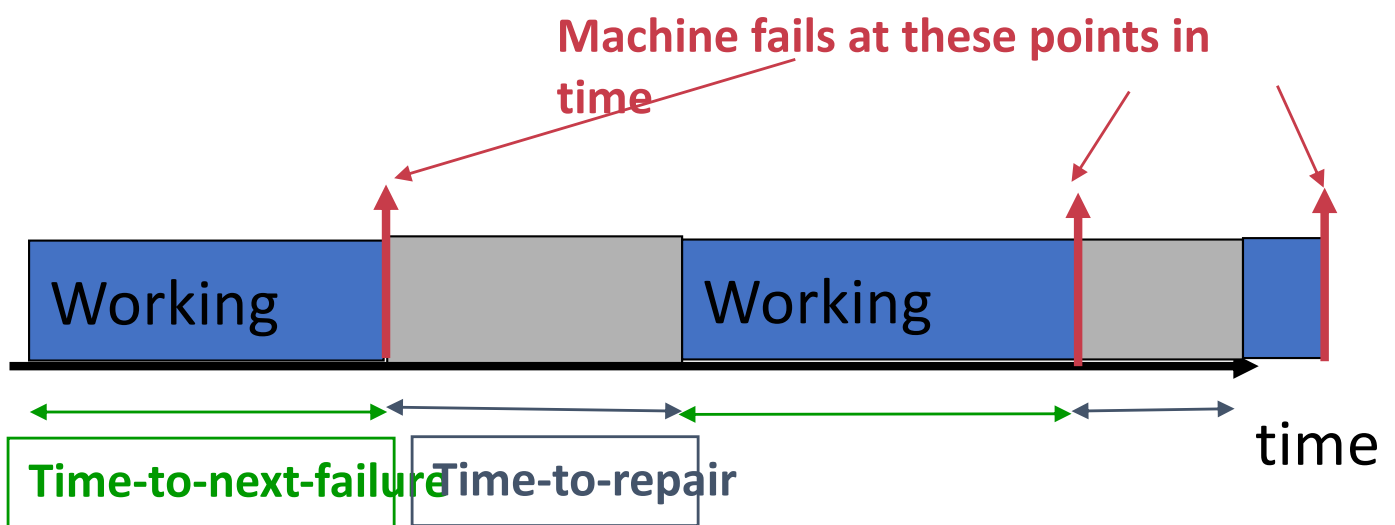
(3,1,0), (3,0,1)

(2,2,0), (2,1,1) (2,0,2)

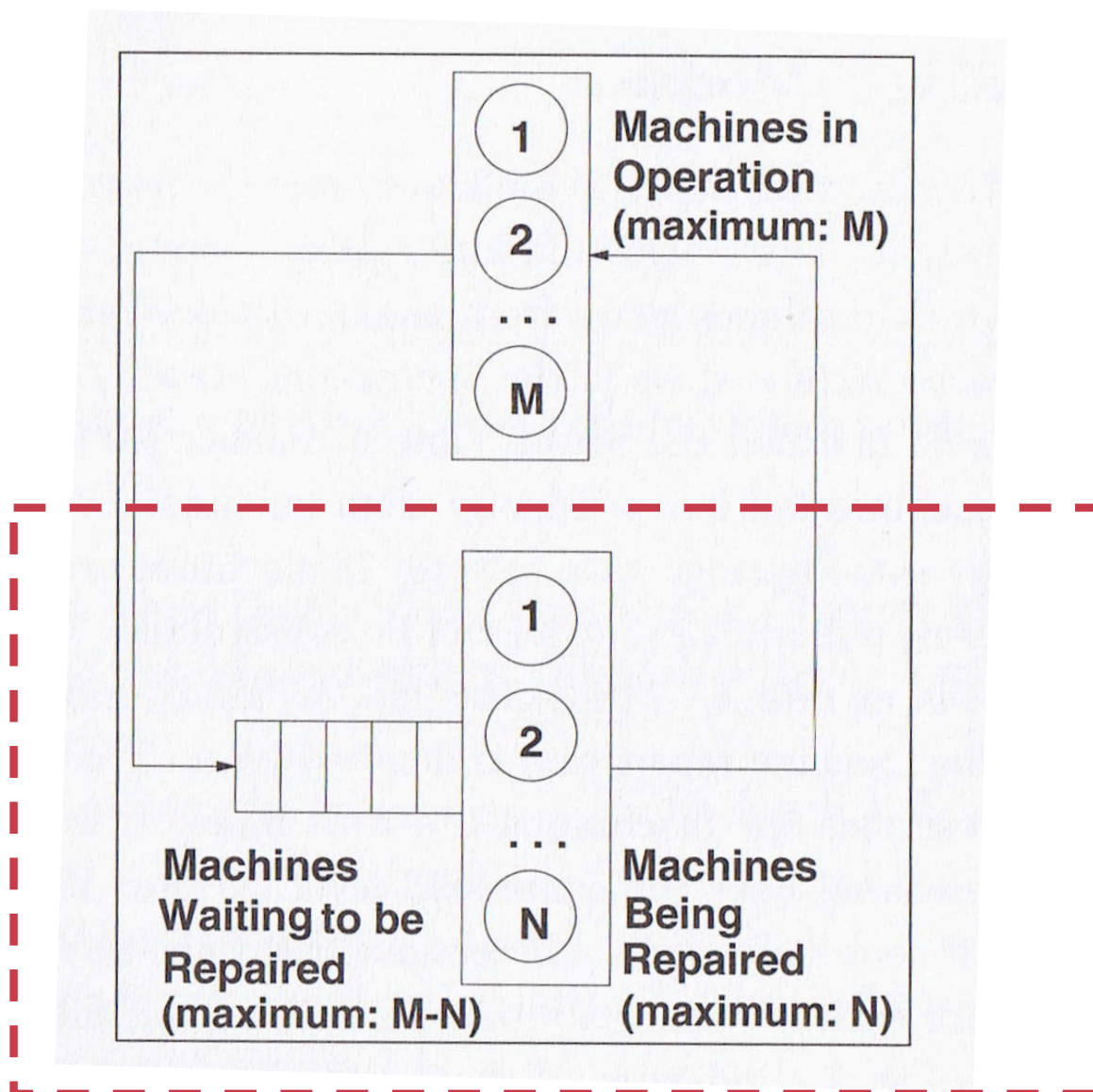
(1,3,0) (1,2,1) (1,1,2) (1,0,3)

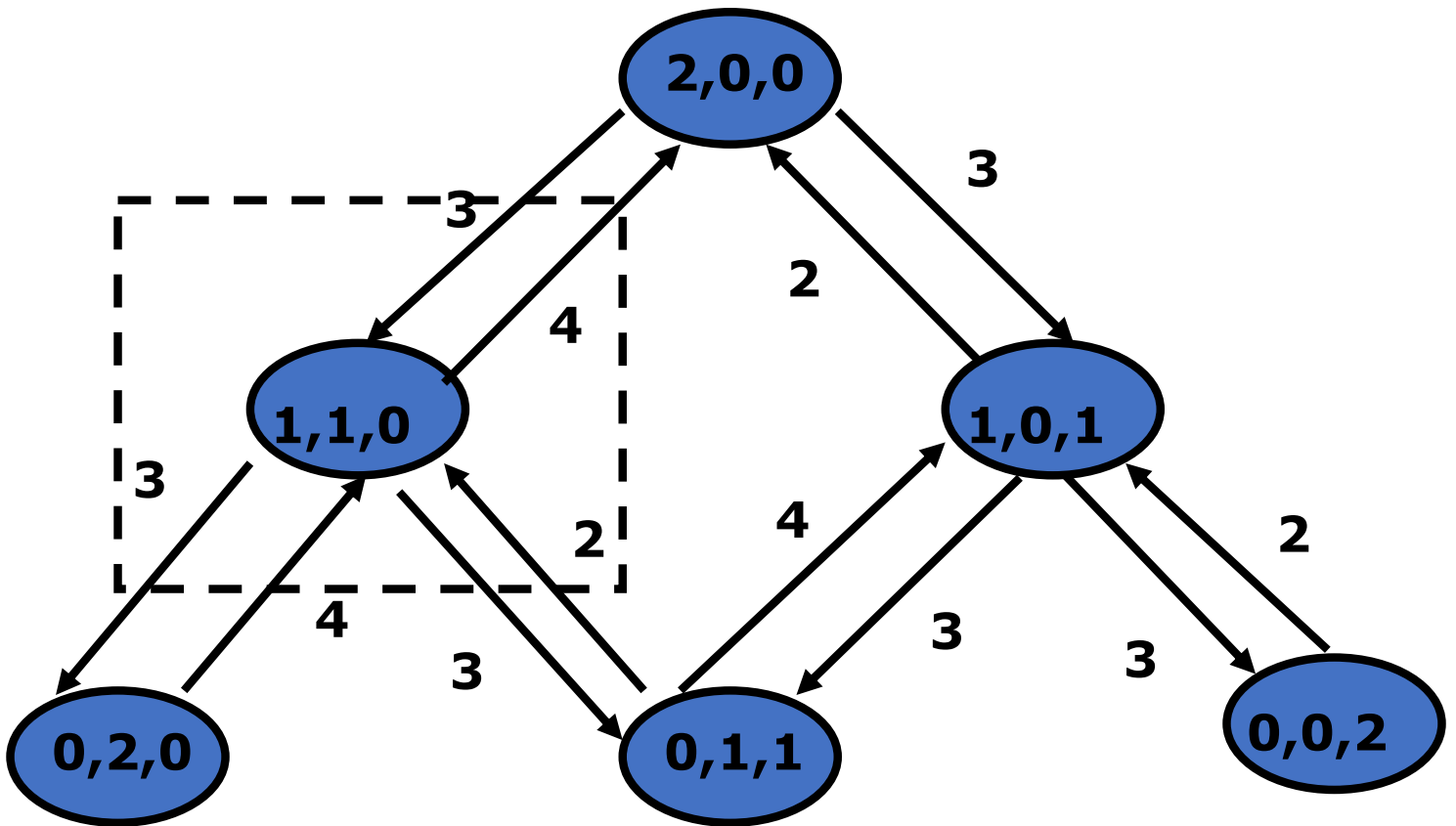
(0,4,0) (0,3,1) (0,2,2) (0,1,3) (0,0,4)

	Meaning
λ	1 / (Mean-time-to-failure for a machine)
μ	1 / (Mean service time to repair a machine)
M	Number of machines ($M > N$)
N	Number of repair staff



	Meaning
λ	1 / (Mean-time-to-failure for a machine)
μ	1 / (Mean service time to repair a machine)
M	Number of machines
N	Number of repair staff
P(k)	Prob[k machines have failed]





Rate of going into the state $(1,1,0)$
 $= 3 P_{200} + 4 P_{020} + 2 P_{011}$

Rate of leaving the state $(1,1,0)$
 $= 4 P_{110} + 3 P_{110} + 3 P_{110}$
 $= 10 P_{110}$

$$10 P_{110} = 3 P_{200} + 4 P_{020} + 2 P_{011}$$